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- Discussion Prompt: Week 2 Camera Calibration
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Your grade: 100%

Your best: 100% · Your highest: 100%

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2.5 Simple Stereo Self-Check Quiz

1. Which of the following are similarity metrics that can be used in template matching?

1 / 1 point

☐ Minimum Sum of Absolute Differences

☐ Minimum Sum of Squared Differences

☐ Maximum Normalized Cross-Correlation

☒ All of the Above

COOCN

Help me practice

Let's chat

Assignment details

All of these methods (Minimum Sum of Absolute Differences, Minimum Sum of Squared Differences, Maximum Normalized Cross-Correlation) are used in the process of template matching.

Nov 13, 12:41 PM CST · Updated

Try again

2. Detailed structure of an object can be recovered from a Single View. True or False?

1 / 1 point

☐ True

☒ False

100%

View submission

See feedback

Correct

Detailed structure recovery of an object from a single view is not possible. While certain basic aspects of an object's structure and depth information can be inferred from a single view, recovering intricate and detailed 3D structure is typically **difficult** or **impossible**. Single-view methods often lack the necessary information to capture fine details accurately.

3. How does a stereo camera simulate depth perception in images?

1 / 1 point

☐ By using a large aperture for shallow depth of field

☐ By using a wide-angle lens for a broader field of view

☒ By capturing two slightly different offset images and presenting them separately to each eye

☐ By using digital zoom to enhance the perception of depth

Correct

The primary advantage of stereo imaging in cameras compared to traditional single-lens photography is **enhanced depth perception and realism**. Stereo imaging captures images with depth information, allowing viewers to perceive objects and scenes in three dimensions. This adds a greater sense of realism and spatial understanding compared to flat, single-lens images.

4. What is the main goal of stereo matching in computer vision?

1 / 1 point

☐ To identify the color distribution of images

☐ To match images with similar lighting conditions

☒ To find corresponding points in multiple images for depth perception

☐ To detect and remove image noise

Correct

The main goal of stereo matching in computer vision is to find corresponding points in two or more images captured from different viewpoints. By identifying these corresponding points, we can calculate the **disparity** (horizontal pixel difference) between the points and use this information to estimate the depth or distance to objects in the scene.

https://www.coursera.org/learn/3d-reconstruction-multiple-viewpoints/assignment/submission/nbv2-5-simple-stereo-self-check-quiz/view-feedback

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